Rizki Perdana

Full-Stack Software Engineer | Product, Cloud & Applied AI

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PROFESSIONAL PROFILE

Full-stack software engineer with 8+ years of experience delivering B2B products and web applications. I work across React and Next.js interfaces, Python backend services, relational data, cloud platforms and infrastructure as code. My applied-AI work covers model selection and evaluation, prompt engineering, document extraction, RAG/retrieval, agentic workflows, MCP integrations and controlled tool use. I take features from problem definition and system design through implementation, deployment and production support.

PROFESSIONAL EXPERIENCE

SourceSage | **March 2022 - Present** | **Singapore**
Senior Software Engineer

- Own full-stack delivery across a B2B procurement platform, building Python/Flask services and React interfaces backed by SQLAlchemy, MySQL, Celery and external integrations.
- Design supplier, order, payment and document workflows across API contracts, relational data models, asynchronous processing and user-facing product features.
- Lead an AI-assisted purchase-order workflow: select and evaluate LLM and vision models, engineer prompts and extraction/normalization logic for email and PDF orders, and retain human approval before order creation and invoicing.
- Reduced the purchase-order extraction error rate from roughly 50% initially to around 3% after model and workflow iteration; outputs remain human-reviewed.
- Build agentic workflows and MCP tool integrations, applying RAG/retrieval, context design, tool calling and agent-harness patterns with explicit read-only and approval boundaries.
- Manage cloud environments across AWS and GCP; handle deployment configuration, service and instance operations, Terraform-based infrastructure changes and production releases.
- Mentor junior engineers and contribute to solution design, code review, technical decisions and production support.

Binar Academy | **February 2020 - February 2022** | **Jakarta, Indonesia**
Software Engineer

- Delivered client-facing web features and REST APIs, contributing to database design, frontend implementation, service integrations, testing and releases.

PT Gamatechno Indonesia | **February 2018 - February 2020** | **Yogyakarta, Indonesia**
Software Engineer

- Developed and maintained web applications and REST APIs, including authentication and data-handling features; supported deployment and application operations.

SELECTED PROJECTS

semgrepll | Local semantic code search for AI coding agents, combining local embedding/model runtimes with SQLite and LanceDB vector retrieval. PyPI.

Lazy Matcher | Asynchronous job-matching pipeline using FastAPI, PostgreSQL, Redis workers, LLM scoring and deterministic fallbacks.

WikWok | Mobile-first Wikipedia discovery product built with Next.js, React and TypeScript, featuring search, multilingual content and an installable PWA. Live demo.

TECHNICAL EXPERTISE

Frontend & product: React, Next.js, JavaScript, TypeScript, responsive UI, component-driven development

Backend & APIs: Python, Flask, FastAPI, SQLAlchemy, Celery, REST APIs, asynchronous processing, integrations

Cloud & infrastructure: AWS, GCP, Terraform, Docker, serverless deployments, CI/CD, infrastructure configuration and service operations

Data & search: MySQL, PostgreSQL, Redis, Elasticsearch, OpenSearch, vector search and retrieval

AI & agent engineering: LLM and vision-model selection/evaluation, inference and embeddings, prompt engineering, document extraction, RAG/retrieval, agentic workflows, MCP, tool calling, agent harnesses and human-in-the-loop controls

EDUCATION & LANGUAGES

Yogyakarta State University | Bachelor of Accounting | 2016 - 2020 | **Languages:** Indonesian (native), English (professional working proficiency)